CHAPTER 6

Boolean Functions: SOP, POS, Minterms & Maxterms

Truth table ↔ expression · minterm & maxterm · Σ and Π notation · canonical forms · don't-cares

QUICK TRICK
Main idea of this chapter: read the **1s** of a truth table to get SOP. Read the **0s** (and flip) to get POS. The tricks are in the green boxes.

6.1 What is a Boolean function?

● **LEVEL 1 · BEGINNER**

What is it? A Boolean function gives **one output (0 or 1)** for every possible combination of inputs. Example: F(A, B, C) means the output F depends on three inputs A, B and C.

Three ways to show the same function:

Way	Example
Truth table	A list of all input combinations and the output for each
Boolean expression	F = A·B' + C
Logic circuit	AND, OR and NOT gates joined together

Why do we need to switch between them? A problem is usually given in words or as a truth table. We must turn it into an expression, simplify it, and then build the circuit.

EASY MEMORY
n inputs → 2^n rows in the truth table. 2 inputs = 4 rows, 3 inputs = 8 rows, 4 inputs = 16 rows.

Expression → Truth table

Steps: write all input combinations (count in binary: 000, 001, 010 …), work out each part of the expression, then combine. Example: **F = A·B' + C**

A	B	C	B'	A·B'	F = A·B' + C
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

QUICK TRICK

Quick method: do not fill every column. Find where each product term is 1. A·B' is 1 only when A = 1 and B = 0 (rows 100, 101). C is 1 whenever C = 1 (rows 001, 011, 101, 111). F is 1 on **all rows where at least one term is 1**: rows 1, 3, 4, 5, 7.

6.2 SOP and POS

● **LEVEL 1 · BEGINNER**

	SOP — Sum of Products	POS — Product of Sums
Meaning	An **OR** of **AND** terms	An **AND** of **OR** terms
Example	AB + A'C + BC'	(A + B)(A' + C)(B + C')
Shape of circuit	AND gates → one OR gate	OR gates → one AND gate
You read it from the truth table	From the rows where F = **1**	From the rows where F = **0**

EASY MEMORY

Sum = OR (+). Product = AND (·). "Sum of products" = OR of ANDs. "Product of sums" = AND of ORs.

COMMON MISTAKE

AB + AC is SOP. **(A + B)(A + C)** is POS. But **A + BC** is also SOP (A is a product with one variable). Do not call it POS just because it has brackets somewhere.

6.3 Minterm and Maxterm

● **LEVEL 2 · EXAM**

Minterm: an **AND** term that contains **every** variable (plain or with a bar). It equals 1 on **exactly one** row.

Maxterm: an **OR** term that contains **every** variable. It equals 0 on **exactly one** row.

Every row gets a number (the row number in binary). Minterm of row i is written $\mathbf{m_i}$ and maxterm $\mathbf{M_i}$.

QUICK TRICK

Minterm rule: 1 → plain letter, 0 → letter with bar. (Row 101 → A B' C.)
Maxterm rule: 0 → plain letter, 1 → letter with bar. (Row 101 → A' + B + C'.)
Maxterm is the opposite of minterm. $\mathbf{M_i = (m_i)'}$.

Row	A B C	Minterm	Maxterm
0	000	m_0 = A'B'C'	M_0 = A + B + C
1	001	m_1 = A'B'C	M_1 = A + B + C'
2	010	m_2 = A'BC'	M_2 = A + B' + C
3	011	m_3 = A'BC	M_3 = A + B' + C'
4	100	m_4 = AB'C'	M_4 = A' + B + C
5	101	m_5 = AB'C	M_5 = A' + B + C'
6	110	m_6 = ABC'	M_6 = A' + B' + C
7	111	m_7 = ABC	M_7 = A' + B' + C'

EASY MEMORY
minterm = **m**ake it 1 (AND, one row gives 1). **M**axterm = **M**ake it 0 (OR, one row gives 0). Small m → small (AND). Capital M → big (OR).

6.4 Canonical forms and Σ / Π notation

● **LEVEL 2 · EXAM**

Canonical SOP = OR of all minterms where F = 1. **Canonical POS** = AND of all maxterms where F = 0. "Canonical" means *every* term contains *every* variable.

Short notation: **Σm(…)** (sigma) lists the minterm numbers where F = 1. **ΠM(…)** (pi) lists the maxterm numbers where F = 0.

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TRUTH TABLE ---> SOP : take the rows where F = 1.
                       Write one MINTERM for each row. OR them together.
TRUTH TABLE ---> POS : take the rows where F = 0.
                       Write one MAXTERM for each row. AND them together.
```

Worked example: truth table → SOP and POS

Row	A	B	C	F	Term to write
0	0	0	0	0	M_0 = (A + B + C)
1	0	0	1	1	m_1 = A'B'C
2	0	1	0	1	m_2 = A'BC'
3	0	1	1	0	M_3 = (A + B' + C')
4	1	0	0	1	m_4 = AB'C'
5	1	0	1	0	M_5 = (A' + B + C')
6	1	1	0	0	M_6 = (A' + B' + C)
7	1	1	1	1	m_7 = ABC

SOP (rows with F = 1: 1, 2, 4, 7):

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F = SIGMA m(1, 2, 4, 7) = A'B'C + A'BC' + AB'C' + ABC
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POS (rows with F = 0: 0, 3, 5, 6):

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F = PI M(0, 3, 5, 6) = (A+B+C) (A+B'+C') (A'+B+C') (A'+B'+C)
```

Both are the same function. (This F is 1 when an **odd** number of inputs are 1: the 3-input XOR.)

QUICK TRICK
Quick check: number of minterms + number of maxterms = 2^n. Here 4 + 4 = 8. ✔

6.5 Converting between forms

● LEVEL 2 · EXAM

QUICK TRICK

Σ ↔ Π (the "rest" trick): Σm(list) = ΠM(all the other row numbers). Write all row numbers 0 to $2^n - 1$. Cross out the ones in the list. The rest is the other list.
Example (3 variables): Σm(1, 2, 4, 7) = ΠM(0, 3, 5, 6).

QUICK TRICK

Complement of a function: F' = Σm(the rows where F = 0). So if F = Σm(0, 1, 2, 5), then F' = Σm(3, 4, 6, 7). In Π form: F = ΠM(3, 4, 6, 7) and F' = ΠM(0, 1, 2, 5). **Complement = swap the lists.**

Normal expression → canonical form (add the missing variables)

If a term is missing a variable X, multiply it by **(X + X')** (this equals 1, so nothing changes). Then expand.

Example 1 (SOP): F(A,B,C) = A + B'C

A is missing B and C: A(B + B')(C + C') = ABC + ABC' + AB'C + AB'C' → minterms 7, 6, 5, 4

B'C is missing A: (A + A')B'C = AB'C + A'B'C → minterms 5, 1

Join all (5 appears twice, keep once): **F = Σm(1, 4, 5, 6, 7)**

Check: A = 1 gives rows 4–7 ✔. A = 0 needs B'C = 1, which is row 001 = 1 ✔.

QUICK TRICK

Shortcut with the number of rows: a term with k variables (out of n) covers $\mathbf{2^{n-k}}$ minterms. In 3 variables: A (1 variable) covers $2^2 = 4$ rows. B'C (2 variables) covers $2^1 = 2$ rows. Just list the rows where the term is true.

Example 2 (POS): F = (A + B)(A' + C)

(A + B) is missing C: A + B + CC' = (A + B + C)(A + B + C') → maxterms M0, M1

(A' + C) is missing B: A' + C + BB' = (A' + B + C)(A' + B' + C) → maxterms M4, M6

F = ΠM(0, 1, 4, 6)

Check: F = 0 when A = B = 0 (rows 0, 1) or when A = 1 and C = 0 (rows 4, 6). ✔

Example 3 (Level 3): the majority function

F = 1 when at least two of A, B, C are 1. Rows: 011, 101, 110, 111 → **F = Σm(3, 5, 6, 7)**

F = A'BC + AB'C + ABC' + ABC. Simplify: ABC + A'BC = BC, ABC + AB'C = AC, ABC + ABC' = AB.

(ABC can be used again and again because X + X = X.) **F = AB + BC + AC**.

6.6 Don't-care conditions

● LEVEL 2 · EXAM

What is it? Sometimes an input combination **never happens**, or we do not care about the output for it. We mark that output with **X** (or d). **Why useful?** We may choose X to be 0 or 1, whichever makes the expression **simpler**.

Notation: F = Σm(1, 3, 7) + d(5) means F is 1 for rows 1, 3, 7, and we do not care about row 5.

Example: BCD even-digit detector

A BCD digit uses 4 bits (A B C D), so only 0 to 9 appear. Rows 10 to 15 never occur.

F = 1 for even digits: **F = Σm(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)**

Choose X = 1 for rows 10, 12, 14 (they are even). Now every even row is 1 and every odd digit (1, 3, 5, 7, 9) is 0.

The simple answer is **F = D'** (the last bit is 0 for even numbers). One NOT gate! Without don't-cares it would need many gates.

EXAM TRAP

Don't-cares are **not** written in the SOP or POS list as real minterms. Keep them in a separate d(...) list. In the final circuit each X becomes either 0 or 1 (depending on how you used it).

6.7 Counting facts that GATE likes

● **LEVEL 3 · GATE**

Fact	Value
Number of minterms for n variables	2^n
Number of different functions of n variables	2^{2^n} (n = 3 → 256, n = 4 → 65536)
Sum of ALL minterms / product of ALL maxterms	1 / 0
If F has m minterms, how many does F' have?	$2^n - m$
A product term with k literals in n variables	covers 2^{n-k} minterms

QUICK TRICK

Minterms inside a term: in 4 variables, the term AB'D' has 3 literals, so it covers $2^{4-3} = 2$ minterms (A = 1, B = 0, D = 0, and C can be 0 or 1): m8 (1000) and m10 (1010).

Quick-Solve Tricks — Summary

Situation	Trick
Make SOP from a truth table	Take the rows where F = 1. Minterm: 1 → plain, 0 → bar. OR them.
Make POS from a truth table	Take the rows where F = 0. Maxterm: 0 → plain, 1 → bar. AND them.
Switch Σ ↔ Π	Use “the rest” of the row numbers.
Complement of F	Swap the Σ and Π lists (zeros become ones).
Missing variable in a term	Multiply by (X + X') and expand. Or list the rows where the term is true.
Rows covered by a term	2^{n-k} rows for k literals.
Don't care	Choose 0 or 1 to make a bigger/simpler group.

GATE QUICK CHECK — Chapter 6

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Minterm  m_i : AND of all variables. 1 -> plain, 0 -> bar.  Equals 1 on ONE row.
Maxterm  M_i : OR  of all variables. 0 -> plain, 1 -> bar.  Equals 0 on ONE row.
M_i = (m_i)'      SIGMA m(list) = PI M(rest)       F' : swap the lists
Rows = 2^n        Functions = 2^(2^n)               k literals cover 2^(n-k) rows
```

Traps: using the wrong bar rule for maxterms · reading the 1s when asked for POS · forgetting the missing variable · putting don't-cares in the Σ list.

Practice questions

● LEVEL 1 · BEGINNER

Q1. How many rows are there in the truth table of a 4-variable function?

Q2. Write the minterm m_5 for variables A, B, C.

Q3. Write the maxterm M_3 for variables A, B, C.

● LEVEL 2 · EXAM

Q4. Convert F(A,B,C) = Σm(1, 2, 5) to Π form.

Q5. Write F = AB + C' as Σm (three variables A, B, C).

Q6. A function is 1 only on rows 3, 5, 6, 7 (three variables). Write the canonical SOP and the simplified SOP.

● LEVEL 3 · GATE

Q7. In 4 variables (A, B, C, D), which minterms does the product term AB'D' contain?

Q8. F(A,B,C) = Σm(0, 1, 2, 5). Write F' as Σm and F as ΠM.

Q9. F(A,B,C) = ΠM(0, 3, 5, 6). Write F as Σm.

Q10. How many different Boolean functions exist for 4 variables?

Q11. F = Σm(1, 3, 7) + d(5) for A, B, C. Find the simplest expression.

Q12. F(A,B,C,D) = Σm(0, 2, 4, 6, 8, 10, 12, 14). Simplify.

Answers

Q	Solution
1	2^4 = **16** rows.
2	5 = 101 → 1 plain, 0 bar, 1 plain → **AB'C**.
3	3 = 011 → 0 plain, 1 bar, 1 bar → **A + B' + C'**.
4	All rows 0–7. Remove 1, 2, 5 → **ΠM(0, 3, 4, 6, 7)**.
5	AB is true on rows 110, 111 → 6, 7. C' is true on rows 000, 010, 100, 110 → 0, 2, 4, 6. Join → **Σm(0, 2, 4, 6, 7)**.
6	Canonical: **A'BC + AB'C + ABC' + ABC**. Simplified (majority): **AB + BC + AC**.
7	A = 1, B = 0, D = 0, C free → 1000 and 1010 → **m8 and m10** (2 minterms; 2^{4-3} = 2).
8	F' has the other rows: **F' = Σm(3, 4, 6, 7)**. F = **ΠM(3, 4, 6, 7)**.
9	Rest of {0–7} without 0, 3, 5, 6 is 1, 2, 4, 7 → **Σm(1, 2, 4, 7)**.
10	$2^{2^4} = 2^{16}$ = **65536**.
11	Take X (row 5) as 1: rows 1, 3, 5, 7 all have C = 1 → **F = C**.
12	All even rows have D = 0 and there are 8 of them (all rows with D = 0) → **F = D'**.

Next: Chapter 7 — Karnaugh Map (K-map): grouping, prime implicants, SOP and POS minimisation, don't-cares, with quick tricks.